

Bio-transforming maize husk residue into a multi-omics-validated biostimulant: Agro-waste bioformulation sustaining micronutrient delivery in millets

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Highlights

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Maize husk extract (MHE) delivers Fe and Zn, turning agro-waste into a sustainable biostimulant.

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Multi-omics reveals MHE is rich in chelators and boosts beneficial, nutrient-mobilizing microbes.

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Fe-Zn incubation with MHE forms nanoscale, stable complexes, enhancing micronutrient loading and bioavailability.

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MHE offers a low-cost, eco-friendly alternative to synthetic chelates.

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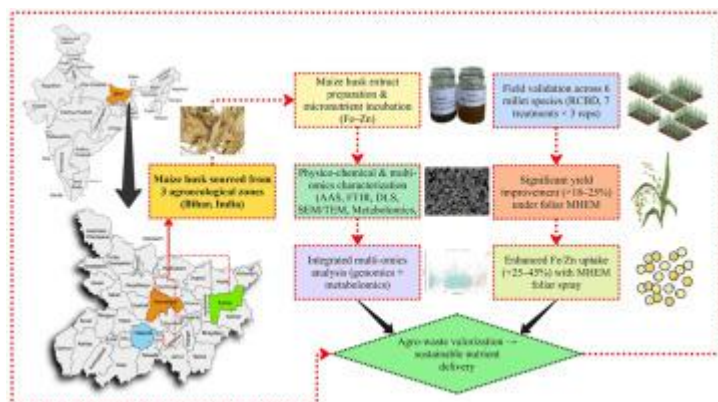
This approach links waste valorisation with efficient micronutrient delivery for sustainable crops.

Abstract

Widespread iron and zinc deficiencies in cultivated soils impair crop performance and nutrition, while agro-waste recycling offers a cost-effective and bioavailable alternative to conventional fertilizers. Present study explored the aspect of transforming maize husk

derived extract as a bioactive material for stabilizing Fe and Zn using a validated multi-omics approach to transform maize waste in bio-input. Maize husk extract incubation with 2.5 % Fe–Zn solution led to reduced levels of free elements (Fe: 11,250 mg L⁻¹; Zn: 12,300 mg L⁻¹) presenting successful micronutrient complexation in Atomic Absorption Spectroscopy. Dynamic Light Scattering revealed that, against the commercial micronutrient (~2000 nm), MHE incubation reduced particle size to ~300–600 nm with improved uniformity. Scanning Electron and Transmission Electron Microscopy further revealed uniform spherical particles embedded within the maize husk, confirming successful micronutrient stabilization. Fourier Transform Infrared Spectroscopy analysis indicated intensified chelating functional groups (–OH, –COOH), and metabolomic profiling showed ~2-fold enrichment of citric (RT ~12.3 min), malic (RT ~15.6 min), and gallic acids (RT ~21.4 min), acting as key chelators. Metagenomic analysis identified siderophore-producing *Streptomyces scabrisporus* (0.31 %) and *Pseudomonas umsongensis* (0.52 %) as dominant contributors to organic acid biosynthesis and micronutrient mobilization. Subsequent field trials with six millet crops confirmed that MHE-based foliar formulations significantly enhanced Fe and Zn uptake (25–45 %) and increased grain yield by up to 25 %, especially in foxtail and pearl millet, validating the bioavailability and agronomic efficacy of this waste-derived innovation. These findings validate MHE as an ecofriendly biostimulant that improves micronutrient bioavailability while valorizing agro-waste for sustainable agriculture.

Graphical abstract



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Introduction

Iron and zinc deficiencies in soils are major constraints to global agriculture, impairing crop productivity, nutritional quality, and food security. These micronutrients are vital for plant

physiological processes such as photosynthesis, enzyme activation, and stress tolerance, and their deficiency contributes to yield losses of 10–25 % in affected regions [1]. Globally, about 30 % of arable land is low in iron and zinc. In India, this figure rises to around 50 %. This deficiency leads to yield losses of 10 %–25 % and affects the nutrition of over 800 million people [2]. These deficiencies lead to abiotic stresses and thus double the impact on food security and farmer livelihoods. Intensive agriculture, unsustainable farming practices like overuse of chemical fertilizers, and depletion of soil organic matter have reduced the availability of these micronutrients in soil [3]. Conventional fertilization methods suffer from low uptake efficiency (30–50 %) due to leaching and soil fixation, leading to high costs and ecological pollution [4,5]. Increased production costs and nutrient runoff leading to ecological pollution trigger the need of sustainable alternatives that enhance micronutrient bioavailability. By incubating micronutrients with plant extracts, their bioavailability is enhanced due to natural chelation by organic acids and phenolics, which reduces soil fixation and improves plant responses [6,7]. Products derived from such extracts can boost the efficiency of micronutrient usage by 30–40 % while also fostering beneficial soil microbes, presenting an environmentally friendly solution for crop nutrition [8].

Agricultural waste valorisation can potentially address challenges of soil micronutrient deficiencies. Although a large amount of agricultural waste is produced, their handling is still inefficient with much of it being burned or allowed to rot, causing pollution and greenhouse gases emissions. Maize husk is an example of underutilized agricultural waste besides being an important byproduct of maize cultivation. Open-field decomposition of maize husk is a known source of methane and other greenhouse gases, intensifying the challenges of climate change [9]. Synthetic micronutrient products are costly, poorly bioavailable, and environmentally flawed due to excessive nutrient loss. Despite some advances, agro-waste-derived extracts remain under-validated as micronutrient carriers, particularly at nano or conventional scales. Their use in dryland cereals like millets, often grown in micronutrient-deficient soils is rare. Prior studies have focused on nano-chelates and synthetic formulations, but their high cost and ecological risks limit farmer adoption.

Increasing iron and zinc deficiencies and the declining consumption of millet is a pressing concern in this century [10]. Millet is emphasized due to its significant potential to address these deficiencies. Although millets' nutritional richness and climate resilience are increasingly acknowledged, iron and zinc deficits frequently limit their productivity and grain micronutrient density. Biofortification with FeO/ZnO nanoscale particles has shown notable yield and nutrient improvements in cereals like rice [11], while recent developments employing fortified nanoformulations have showed encouraging results in vegetables like cabbage and cauliflower [12]. Dryland cereals like millets are essential for

nutritional security in marginal areas while being rarely examined for such sophisticated nutrient systems. Commercial nanoformulations and synthetic chelates provide targeted distribution, but they are frequently too expensive and environmentally harmful for widespread use in small and marginal farms.

Maize Husk Extract (MHE) holds strong potential to foster beneficial microbial communities, a claim supported by metagenomic studies. Globally, millions of tons of maize husk are generated each year [13]. In India, particularly in Bihar, maize is cultivated across three seasons, resulting in substantial maize residue. While some of this residue is used for fuel and livestock feed, a considerable portion remains unutilized [14]. Rich in phenolics, flavonoids, and organic acids, maize husk is often discarded contributing to waste accumulation and greenhouse gas emissions during decomposition [15].

The transformation of maize husk into a useful biostimulant is therefore possible, where waste can be reduced and productivity in agriculture improved. Bioactive metabolites extracted from maize husk have been shown to improve the solubility, mobility, and uptake of iron and zinc through chelation and particle size reduction. MHE potentially interact with micronutrients at the molecular level, forming stable complexes that prevent soil fixation and facilitate transport to plant tissues. Additionally, microbes play synergistic role in nutrient cycling through siderophore production, organic acid secretion, and other metabolic pathways, amplifying nutrient uptake efficiency.

This study explores the transformation of maize husk into a multi-omics-validated MHE as a biogenic carrier for iron and zinc delivery and influence on nutrient bioavailability. We hypothesised that MHE incubated with iron (Fe) and zinc (Zn) could create stable bioactive complexes that improve the bioavailability of micronutrients through combined metabolomic and microbial mechanisms. To fulfil the objective, research was carried out in two phases: (i) produced and characterized a micronutrient-incubated MHE utilizing multi-omics methods, and (ii) field trial with six millets to evaluated its effectiveness in increasing Fe and Zn uptake under actual field conditions.

Maize husk collection and processing

The maize husk used in the extraction was collected from three distinct agroecological zones in Bihar, India: Samastipur (Zone I), Purnia (Zone II), and Nalanda (Zone III) during the kharif season (varieties DKC 7074, P 3378, and DKC 9144). Sampled crops were cultivated following standard agronomic practices while no foliar micronutrients were applied. At physiological maturity, maize husks were collected manually and processed to reduce variability. The composite husk samples were dried for 5

Nutrient composition (Zn and Fe)

The AAS analysis revealed little changes in iron (Fe) and zinc (Zn) concentrations upon incubating the commercial micronutrient solution with MHE (Supplementary file ST-1). Initially, the commercial solution contained Fe and Zn at concentrations of 25,000 mg L⁻¹ each (2.5 %). In contrast, MHE exhibited relatively low inherent concentrations of these nutrients, with only 5.67 mg L⁻¹ of Fe and 8.32 mg L⁻¹ of Zn. However, MHEM with the commercial micronutrient solution in a 1:1 (v/v) ratio, the

Composition and surface properties

The present investigation highlighted the significance of MHE in enhancing Zn and Fe bioavailability, persistent challenges in micronutrient management for sustainable agriculture. The chemical transformations induced during micronutrient incubation as confirmed with FTIR spectroscopy revealed spectral bands indicative of alcohols, phenolics, and organic acids [20]. The spectral peaks in MHEM at 452.9 cm⁻¹ and a shift of the 411.9 cm⁻¹ band to 430.5 cm⁻¹ suggested a reorganization of functional

Conclusion

This research highlights the promise of maize husk extract (MHE) as an eco-friendly biogenic carrier that boosts the bioavailability of Fe and Zn while also making use of agricultural waste. Incubation with micronutrients produced nanoscale complexes stabilized by organic acids (citric, malic, gallic) and functional groups (–OH, –COOH), confirmed through FTIR, TEM, and metabolomic profiling. Metagenomic analysis revealed enrichment of siderophore-producing taxa and organic acid biosynthetic

CRedit authorship contribution statement

Shubham Anil Durgude: Conceptualization, Methodology, Investigation, Writing – original draft, Data curation, Formal analysis, Visualization. **Raj Kumar Jat:** Supervision, Project administration, Conceptualization, Writing – review & editing. **Vijay Singh Meena:** Methodology, Supervision, Writing – review & editing. **Manish Kumar Vishwakarma:** Writing – review & editing. **Shiv Vendra Singh:** Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Suneel Kumar:** Writing –

Funding

The International Maize and Wheat Improvement Center, Borlaug Institute for South Asia, Pusa, Samastipur is sincerely acknowledged by all the authors for providing resources needed for this research. Authors extend sincere appreciation to field staff as well for facilitating this study.